

Capstone Project Report

Smart Attendance System Using Facial Recognition

Problem Statement

Manual attendance marking in classrooms and lecture halls is time-consuming, prone to proxy attendance, and takes several minutes away from every session. Existing RFID or fingerprint-based systems require physical contact at a single entry point, which does not scale well to large lecture halls with multiple doors. This project builds a contactless, camera-based attendance system that enrolls each student's face once, identifies enrolled students automatically as they enter, logs attendance in real time within a defined class session, flags irregularities such as duplicate or unrecognized check-ins separately from confirmed records, and gives faculty a dashboard to review those flags before they reach the official attendance record.

Objectives

- Build a facial-recognition pipeline that reliably identifies enrolled students from a live camera feed under normal classroom lighting.
- Build an enrollment step that registers each student's face once and stores it for later matching.
- Build an attendance database that updates in real time and prevents duplicate check-ins within the same class session.
- Detect and flag unrecognized or duplicate check-in attempts separately from confirmed attendance records, so faculty can review them.
- Provide a faculty-facing dashboard that surfaces low attendance and flagged irregularities at the end of each session.
- Evaluate the system's accuracy against a manually verified attendance sheet over a two-week pilot.

Literature Survey

AdaFace and the broader survey of embedding-based recognition methods show that recognition accuracy is now high enough for unconstrained classroom lighting, while real-time detectors such as YOLOv7 and the more recent open-vocabulary YOLO-World confirm that detection latency is no longer the bottleneck for live video pipelines. These general-purpose methods, however, are tuned for accuracy on server-class hardware rather than the modest, fixed classroom cameras this project targets.

Lightweight architectures such as PocketNet and the more recent EdgeFace close that gap, trading a small amount of accuracy for models compact enough to run on-device, but neither is evaluated against a live, multi-door classroom deployment. None of these five works addresses the classroom-specific operational problem of telling a genuine late arrival apart from a duplicate or spoofed check-in in real time, or of giving faculty a way to review a flagged case instead of silently accepting or rejecting it. That is the specific gap this project is built to close through its Duplicate & Anomaly Check, Session Manager, and Faculty Dashboard stages.

- [1] Y. Kim, W. Park, and S. Shin, "AdaFace: Quality Adaptive Margin for Face Recognition," CVPR, 2022.
- [2] C.-Y. Wang, A. Bochkovski, and H.-Y. M. Liao, "YOLOv7: Trainable Bag-of-Freebies Sets New State-of-the-Art for Real-Time Object Detectors," CVPR, 2023.
- [3] M. Wang and W. Deng, "Deep Face Recognition: A Survey," Neurocomputing, 2021.
- [4] T. Cheng, L. Song, Y. Ge, W. Liu, X. Wang, and Y. Shan, "YOLO-World: Real-Time Open-Vocabulary Object Detection," CVPR, 2024.
- [5] F. Boutros, N. Damer, F. Kirchbuchner, and A. Kuijper, "PocketNet: Extreme Lightweight Face Recognition Network Using Neural Architecture Search and Multistep Knowledge Distillation," IEEE Access, 2022.
- [6] A. George, C. Ecabert, H. O. Shahreza, K. Kotwal, and S. Marcel, "EdgeFace: Efficient Face Recognition Model for Edge Devices," IEEE Transactions on Biometrics, Behavior, and Identity Science, 2024.

System Architecture

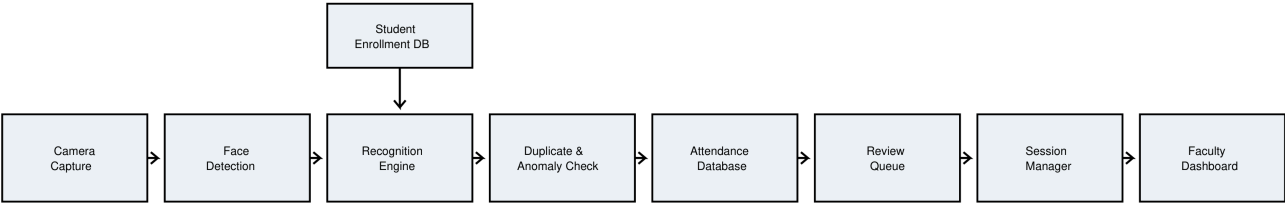
The proposed solution is a nine-component real-time pipeline. Before the system goes live, a student enrollment database registers each student's face embedding once. A camera capture stage continuously grabs frames from a fixed classroom camera. A face detection stage locates face regions in each frame and passes cropped faces to the recognition engine, which compares each detected face's embedding against the enrollment database using cosine similarity and returns a matched student ID above a confidence threshold. A duplicate and anomaly check stage inspects every matched ID against the current session's log: confirmed, non-duplicate matches are written to the attendance database with a timestamp, while low-confidence, unmatched, or repeat check-ins are routed to a review queue instead. A session manager defines each class session's start and end boundaries, resets the duplicate-check window per session, and closes out the session once it ends. At that point a faculty dashboard reads both the attendance database and the review queue and presents a summary, including any flagged irregularities, for faculty to review. Figure 1 shows these nine components and the data flow between them.

System Modules

1. Student Enrollment Module - registers each student's face embedding once before deployment and stores it in the enrollment database that the Recognition Engine matches against.
2. Camera Capture Module - continuously grabs frames from the classroom IP camera and forwards them to the detection stage at a fixed frame rate.
3. Face Detection Module - locates and crops face regions within each frame using a cascade detector, discarding frames with no detected face.
4. Recognition Engine Module - generates a face embedding for each detected crop and matches it against the enrollment database via cosine similarity.
5. Duplicate & Anomaly Check Module - compares each matched ID against the current session log, sending confirmed matches to the Attendance Database and low-confidence, unmatched, or repeat check-ins to the Review Queue.
6. Attendance Database Module - stores timestamped, de-duplicated, confirmed attendance records and exposes a query API for faculty.
7. Review Queue Module - holds flagged check-in attempts separately from confirmed records until a faculty member reviews them.
8. Session Manager Module - defines each class session's start and end window, resets duplicate-check state per session, and closes out the session so the Faculty Dashboard can summarize it.
9. Faculty Dashboard Module - presents an end-of-session summary of confirmed attendance and flagged irregularities from the Review Queue for faculty to review.

Figure 1. System Architecture

Figure 1. Smart Attendance System - Pipeline Architecture



Main flow: left to right, Camera Capture through Faculty Dashboard. Student Enrollment DB feeds Recognition Engine.